

Knowledge before Industrialization

Evidence from Early Copyright Records

Tancredi Rapone^{1,2}

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¹ Barcelona School of Economics (BSE)

² Universitat Autònoma de Barcelona (UAB)

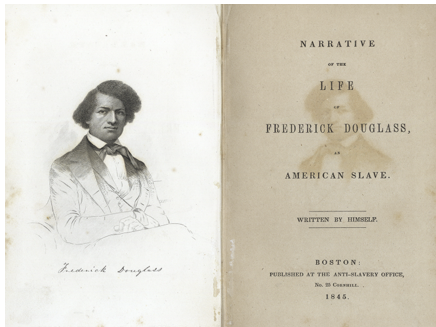
Motivation: knowledge and industrialization

- **Knowledge is central to long-run growth.** Its creation, preservation, and transmission underlie technological change (Mokyr, 2002, 2009).
- **Historical evidence is more limited.** Schooling and literacy describe basic human capital; patents capture inventions.
- Upper-tail human capital (Squicciarini and Voigtländer, 2015) and book production (Baten and van Zanden, 2008) offer complementary windows into the knowledge economy.

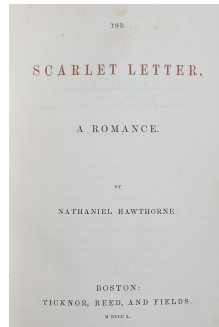
Can we observe the *production of written ideas* before industrialization — and trace the development of the places that produced them?

1. **New data:** Library of Congress copyright title pages, 1790–1870; authors of 1840–1860 works linked to the 1850 census.
2. **Historical pattern:** recorded writing rose sharply before the main late-nineteenth-century industrial expansion.
3. **Local development:** early copyrighting predicts subsequent manufacturing specialization and patenting, conditional on education and baseline county characteristics.
4. **Content:** the relationship with manufacturing is not confined to academic or scientific subjects.

Early copyright records: recognizable and unfamiliar works



Frederick Douglass literary / historical
writing



A second title page original archive
example

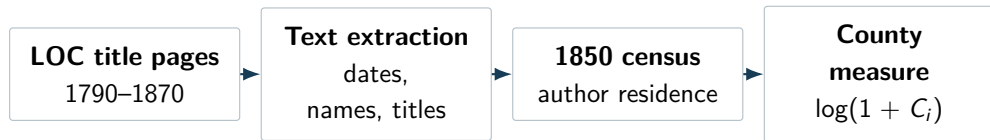
Images from the Library of Congress Early Copyright Title Pages collection, as used in the earlier project slides.

THE
PRIMARY SCHOOL READER.
PART THIRD.
REWRITTEN FOR THE
FIRST CLASS IN PRIMARY SCHOOLS.
AND FOR THE
LOWEST CLASS IN GRAMMAR SCHOOLS.
BY WILLIAM D. SWAN,
PRINCIPAL OF THE MAYHEW GRAMMAR SCHOOL, BOSTON.
BOSTON:
CHARLES C. LITTLE AND JAMES BROWN.
1844.

THE
GENTLEMAN'S
NEW POCKET HARRIER,
CONTAINING A GENERAL DESCRIPTION OF
THE HOUND AND THE BEST METHOD OF
TRAINING HIM;
THE HORSE;
TOGETHER WITH
THE VARIOUS AND ESSENTIAL PARTS OF FURNITURE; NECESSARY TREAT
MENT FOR DOMESTIC ANIMALS; AND A PROPERTY;
HAS CONSIDERATELY AND PRACTICALLY BEING
ADDED FOR IMPROVING THE USE OF A HOUND.
ALSO,
A COMPLETE ACCOUNT OF THE METHODS TO WHICH THE HOUND IS SUBMITTED
WHICH HAVE BEEN RECOMMENDED AS MOST EFFECTUAL AND
ADAPTED TO BE EFFECTUAL.
BY RICHARD MASON, M. D.
SURGEON OF SEVEN OAKS, DISTRICT OF
SOUTH-DORSET, WITH ADDITIONS.
—*—*—*—
TO WHICH IS ADDED,
A PRIZE ESSAY ON MULES;
AND
AN APPENDIX,
Containing Observations and Recipes for the cure of most of the
various distempers incident to Horses, Oxen, Cows,
Calves, Sheep, Swine, Geese, Ducks, &c. &c.
Selected from different authors.
ALSO, AN ADDENDA,
CONTAINING
ANNALS OF THE TURF.
AMERICAN STUD BOOK,
RULES FOR TRAINING, RACING, &c.
PHILADELPHIA:
GRIGG & ELLIOT, No. 3 NORTH FOURTH STREET.
1841.

5/22

A new measure of written production



- Approximately **50,000 works** and more than **90,000 archival images**; about **70,000 dated records** recovered.
- Match registrations in **1840–1860** to authors' county of residence in **1850**.
- **5,162 linked authors**; 71.7% of matching entries have a single eligible census candidate after name, occupation, and age restrictions.

Counts of dated records can include resubmissions and duplicate scans; C_i denotes linked registrations.

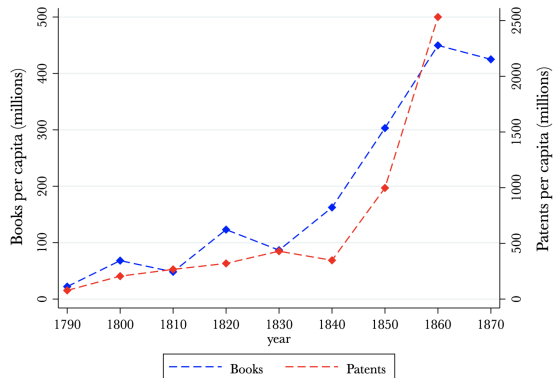
What did people write about?

- The title pages identify subjects as well as authors: teaching, science, professional practice, fiction, religion, and political writing.
- **Academic / non-academic split:** use educational keywords to form initial labels and a logistic text classifier to classify the remaining titles.
- **Academic writing** includes instructional materials, not only original scientific research.

The classification allows us to test whether the local relationship with industry is driven exclusively by formal academic knowledge.

Algorithm and vocabulary in the paper's classification appendix.

Copyright registrations rise before the major industrial expansion



National per-capita copyright and patent series, 1790–1870. Vertical axes have different scales. Source: LOC and U.S.

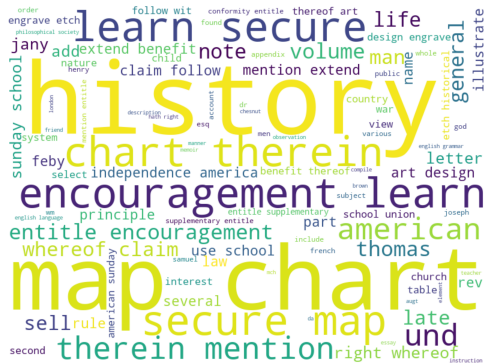
historical patent statistics.

Language of the archive: 1790–1830

1790–1810

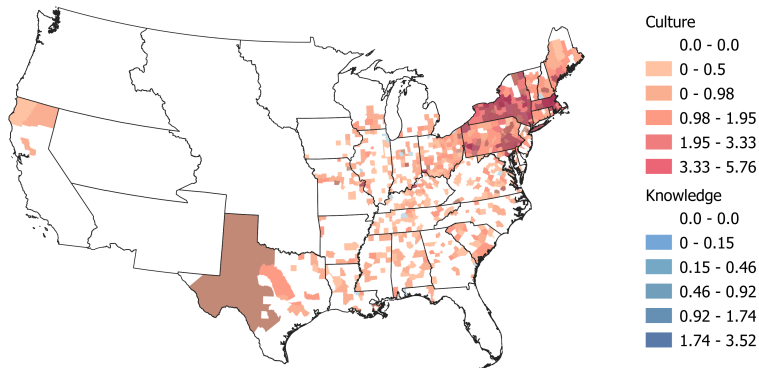


1810-1830



Word clouds from the underlying historical copyright-title-page corpus. Word size reflects frequency in the transcriptions.

Where was written knowledge produced?



Linked registrations, 1840–1860, assigned to authors' counties of residence in 1850. Map distinguishes academic and non-academic content.

Copyrighting counties were different already in 1850

	Above median	Below median
Population	25,797	8,368
Literacy rate	0.876	0.792
Urban population share	0.021	0.000
Number of colleges	0.306	0.053
Schools per child	0.020	0.015

Does copyrighting predict later industry *beyond* these initial differences?

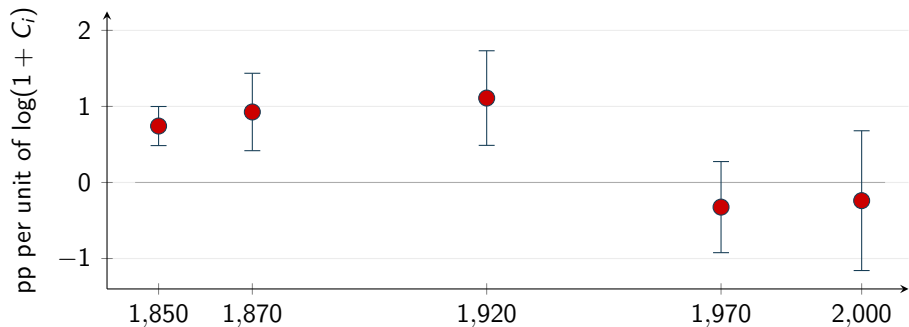
Means across 1,257 counties. Source: descriptive county characteristics in the paper's Appendix Table A1.

$$y_{it} = \alpha_{s(i),t} + \beta_t \log(1 + C_i) + X_i' \gamma_t + P_t(\text{lon}_i, \text{lat}_i) + \varepsilon_{it}$$

- y_{it} : manufacturing employment share at different dates; patenting over 1860–1940.
- X_i : 1850 literacy, school enrollment, urbanization, population, density, and migrant composition, plus geography.
- State fixed effects; cubic latitude–longitude polynomial; spatially robust standard errors (Conley, 100 km).

Separate cross-sections; the 1850 manufacturing comparison is contemporaneous with the registration window.

Early copyrighting predicts manufacturing specialization



Fully controlled OLS estimates and approximate 95% confidence intervals from Conley t statistics. The 1920 estimate is 1.11 pp ($t = 3.50$).

Manufacturing: OLS and migration-based IV

Outcome year	1850	1870	1920	1970	2000
OLS: full controls	.0074	.0093	.0111	-.0032	-.0024
<i>t</i> statistic	5.65	3.57	3.50	-1.06	-0.51
IV: restricted controls	.063	.036	.185	.102	.035
<i>t</i> statistic	16.47	0.44	3.88	1.96	0.81
First-stage KP <i>F</i>	23	13	13	13	13

The strongest selected-year association occurs around 1920.

OLS controls include population and migrant shares; IV retains geography, human capital, and baseline urbanization.

Outcome is manufacturing employment share as a fraction.

Early copyrighting also predicts invention

	OLS Baseline	OLS Full controls	IV Baseline
$\log(1 + C_i)$	.360	.195	2.353
t statistic	7.47	3.40	3.59
Number of counties	1,499	1,498	1,499
First-stage KP F	—	—	14

- Outcome: log patents per capita per decade, 1860–1940.
- The OLS association remains positive when controlling for initial county size, settlement, literacy, and schooling.

OLS baseline and IV exclude the additional population/density/migration controls included in fully controlled OLS.

Instrument: authors moved with the population

	Authors	Other residents
Living outside state of birth, United States	36%	24%
Living outside state of birth, Midwest	78%	45%

- Mid-nineteenth-century settlement drew migrants from origins with different rates of authorship.
- Counties linked to high-authorship origins should have more written production, even conditional on local geography and observed education.

Native-born individuals in the 1850 census. Comparison from the paper's Table 1.

Migration-based variation in authorship

$$Z_i = \sum_{l \in \mathcal{L}, l \neq s(i)} \left(\underbrace{\frac{A_l}{N_l} - \frac{A_{US}}{N_{US}}}_{\text{origin authorship premium}} \right) \underbrace{\pi_{il}}_{\text{county birthplace share}}$$

- A_l/N_l : authors per U.S. resident born in origin l .
- π_{il} : share of county i 's population born in origin l ; destination-state births are excluded.
- Main IV specifications condition on geography, literacy, schooling, urbanization, state effects, and spatial trends.

Exclusion: origin composition must not affect later outcomes through other characteristics of settlers or their destinations.

First-stage strength and the role of settlement controls

Specification	Baseline IV	+ population / migration
Manufacturing, 1920	13	–
Patenting, 1860–1940	14	1

- The instrument predicts local writing conditional on geography, education, and 1850 urbanization.
- Population and migrant controls absorb much of that variation; the resulting IV estimates are imprecise.
- Report **full-control OLS** and the **baseline IV** specifications side by side.

Kleibergen–Paap first stages; the selected-year manufacturing table does not report the full-control IV specification.

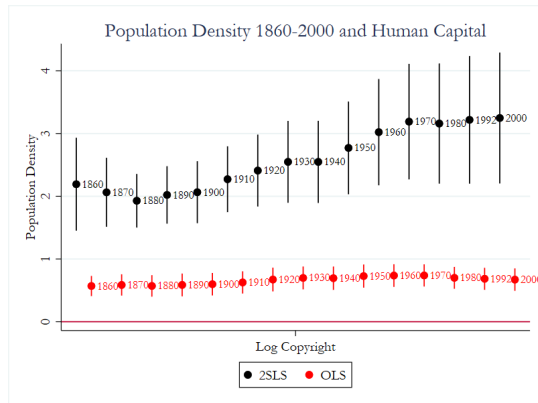
Not only academic writing predicts later manufacturing

	1860	1870	1920
Academic registrations	.00899	.00785	-.00307
<i>t</i> statistic	3.34	2.10	-0.43
Non-academic registrations	.00670	.00873	.01050
<i>t</i> statistic	4.97	3.50	3.25

- Both categories predict manufacturing in 1860–1870.
- By 1920, the statistically significant association remains for **non-academic** writing, conditional on academic writing.

Joint full-control OLS regressions; manufacturing employment share as a fraction. Significance in one category does not by itself test the difference between categories.

Industrial specialization fades; population concentration persists



County population density, 1860–2000: OLS and IV coefficient paths. Unlike the full-control manufacturing OLS regressions, this figure omits population and migrant-composition controls.

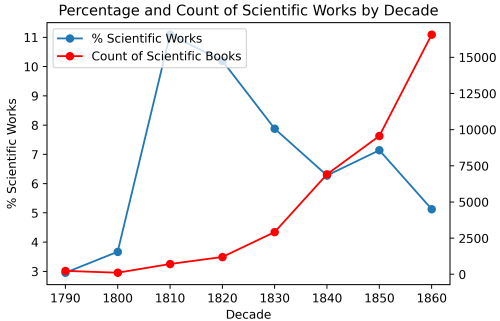
Conclusions

1. **Writing expanded early.** Copyright registrations rose in per-capita terms from around 1830, before the main later industrial expansion.
2. **Intellectual geography anticipated industrial geography.** Counties with more early copyrighting had larger manufacturing shares and more subsequent patents, conditional on baseline characteristics.
3. **The relevant intellectual environment was broad.** Non-academic writing predicts manufacturing as well; the population-density relationship extends into the twentieth century.

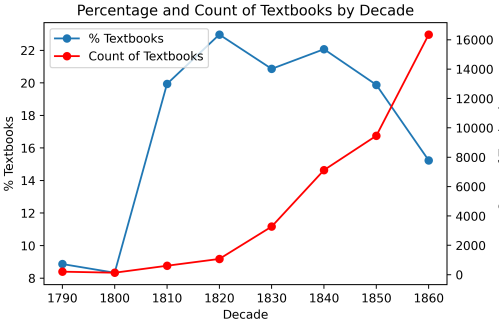
Copyright records provide a new window on the intellectual foundations of American industrialization.

Backup: content of the archive over time

Scientific titles



Textbook-oriented titles



Supplementary descriptive series: topic-specific counts and shares, separate from the paper's binary academic classifier.

Backup: full patenting specification sequence

	(1)	(2)	(3)	(4)	(5)
OLS	.486	.441	.400	.360	.195
<i>t</i>	9.30	12.28	10.55	7.47	3.40
IV	1.448	1.416	1.344	2.353	53.89
<i>t</i>	4.48	4.54	4.51	3.59	0.10
KP <i>F</i>	24	28	22	14	1
Geography	no	yes	yes	yes	yes
Literacy, schools	no	no	yes	yes	yes
Urbanization	no	no	no	yes	yes
Remaining settlement	no	no	no	no	yes

State fixed effects and a cubic latitude–longitude polynomial throughout. Outcome: log patents per capita per decade, 1860–1940.

Backup: geographical balance of the instrument

	(1)	(2)	(3)
Ruggedness	.811	.848	.581
<i>t</i>	3.79	3.18	3.19
Distance to ocean / lake	-.0367	-.0333	-.0102
<i>t</i>	-9.03	-5.55	-1.43
State FE	no	yes	yes
Spatial polynomial	no	no	yes
Counties	1,507	1,507	1,507

Conley 100 km *t* statistics. Other geographic covariates included in paper Appendix Table B1.

Backup: urbanization and income are more sensitive to controls

	OLS, baseline	OLS, full	IV, baseline
Urban share growth, 1860–1940	.0548	.0169	.358
<i>t</i>	3.92	1.11	2.13
Log income per capita, 2000	.0500	.0135	.782
<i>t</i>	4.42	1.04	3.24

- Baseline specifications include geography, schooling, literacy, and initial urbanization.
- Full-control OLS adds population, density, and migrant shares; the estimates become imprecise.

Supplementary results from the paper's Appendix D. IV first stage falls to about 1 with the additional settlement controls.